

8 Design qualification and type tests for Type I insulation systems

8.1 General

There are two stages to the testing of electrical insulation for machines to be fed from converter drives. The first stage is qualification of the materials, insulation system design and manufacturing technique. For Type I insulation systems, it is carried out using motorettes or formettes that undergo thermal cycling and conditioning procedures which include mechanical vibration, moisture exposure and high voltage. Diagnostic tests are performed on these samples and also on complete windings with the aim of assessing the absence of PD. The second stage is a type test performed on the complete winding or machine.

On the basis of these qualification and type test results, a machine is assigned an impulse voltage insulation class, which defines the maximum allowable voltage in units of U_N stressing the parts of the insulation system in converter-fed service (see Annex C).

8.2 Design qualification test

For the purposes of this standard, a qualification test is used to investigate the capability to withstand various stresses. For Type I insulation systems, it is based on PDIV tests before and after thermal cycling and other tests as defined in IEC 60034-18-21 and IEC 60034-18-31, as well as voltage stressing at one of the stress category levels defined in Clause 7, with the voltage increased by the appropriate enhancement factor described in Clause B.3. It is only necessary to perform thermal ageing at any one of the three ageing temperatures specified in IEC 60034-18-21 or IEC 60034-18-31 if the thermal class for the insulation system has already been determined through testing according to these two standards.

8.3 Type test

In the case of Type I insulation systems, PD tests are undertaken to demonstrate the absence of partial discharges [5][6]. The complete winding or machine is subjected to the voltage appropriate to the selected stress category (Table 4), enhanced by a factor (Table B.2). For example, if the application is for a machine which is to be driven by a voltage where the overshoot factor at the terminals is 1,3 (moderate) the stress factors to be used in calculating the test voltage are 1,5 (overshoot) and 0,3 μs (rise time see Clause B.1).

9 Test equipment

9.1 PD measurement at power frequency

A conventional laboratory PD measurement device, using either a high voltage coupling capacitor or a radio frequency current transformer, can be used when the applied voltage to the test object is a sinusoidal 50 Hz or 60 Hz waveform. Details of the test equipment and methods are presented in IEC/TS 60034-27. Use of 50 Hz or 60 Hz applied voltages together with the PD test method described in IEC/TS 60034-27 should be reserved for capacitive test objects such as individual coils, formettes and motorettes.

9.2 PD measurement during voltage impulses

Conventional PD test devices for use with 50 Hz/60 Hz voltages, such as described in IEC/TS 60034-27, cannot generally be used when the applied voltage is a short rise time voltage impulse. A rise time of 0,1 μs has a harmonic content with frequencies of more than 3 MHz. This means that the voltage impulse will have components within the pass band of most IEC/TS 60034-27 style detectors, resulting in a displayed signal that may be hundreds of times the magnitude of the partial discharge pulses. For this reason, it is important to ensure that the partial discharge pulses are distinguished from the high frequency components of the voltage impulses. In addition, the voltage impulses may be of sufficiently high magnitude to destroy the electronics of the partial discharge detector.

To distinguish the partial discharges from the short rise time components of the voltage impulses, a different type of PD detector is needed. The detector should reduce all frequency components from the impulse voltage to less than the high frequency components associated with the partial discharges. The display can be a standard oscilloscope or a pulse magnitude analyser. Guidance is given in IEC/TS 61934 on the method and equipment to be used. It is essential that the PD caused by short rise time impulses are separated from any residue of the impulse voltage. Note that PD remote from the PD sensor may not be detected.

9.3 Voltage impulse generators

Partial discharge tests under impulse voltage conditions require an impulse generator. To simulate properly the impulses for the relevant stress category, the voltage impulse generator should be capable of producing a rise time that is equal to $0,3 \mu\text{s} \pm 0,2 \mu\text{s}$. A larger tolerance may be acceptable for high capacitance windings (see Clause B.2). The impulse generator should have a controllable output magnitude from zero volt to the highest voltage required for the winding voltage rating. When reporting results, the following shall be provided.

- a) The PD sensitivity level, background noise level and detection system noise level as defined in IEC/TS 61934.
- b) The applied impulse voltage under load (test object) and confirmation of compliance with Figures B.1 and B.2.
- c) A photographic or digital record of the impulse waveform at the machine terminals.
- d) The presence or absence of the rotor for tests on complete stator windings.

Reporting of the peak/peak voltage of the impulse generator output and whether it is unipolar or bipolar is advisable.

9.4 Sensitivity

In general, the sensitivity of a partial discharge detector falls as the impedance of the load decreases (or its capacitance increases). As a guide, the sensitivity expected for qualification and type tests covered in this standard should be 1 pC per nF of capacitive load at 50 Hz/60 Hz, with a minimum sensitivity of 1 pC. A measurement system capable of achieving this degree of sensitivity is considered to be sufficiently sensitive to undertake PD tests on inductive loads when equating impedances.

9.5 PD tests

9.5.1 Power frequency voltage

For the purpose of this standard, partial discharge freedom is defined as being less than 5 pC for turn/turn samples and for motorettes and formettes when measurements are made at 50 Hz/60 Hz. These values are also the maximum noise levels permitted during measurements. The PD test is performed according to the procedures in IEC/TS 60034-27.

9.5.2 Impulse excitation

PD tests with impulse voltages are performed according to IEC/TS 61934. The background noise level is given in mV. This background noise level and the sensitivity according to IEC/TS 61934 shall be reported.

10 Qualification of the design of Type I insulation systems

10.1 General

For Type I systems, tests are not carried out to achieve failure by electrical breakdown. The test samples are subjected to thermal and mechanical cycling as described in IEC 60034-18-21 and IEC 60034-18-31 together with various electrical withstand tests. After each sub-cycle,

the test objects are given a partial discharge diagnostic test and the end of test occurs when the PD inception voltage falls below the specified test voltage for the chosen stress category.

Qualification is by comparison with the performance of a reference system as defined in 4.3 of IEC 60034-18-1:2010 which has been qualified under the conditions in 10.4. The candidate shall endure the same, or more, ageing cycles than the reference system without PD occurring below the specified test value. The inception voltage is the lowest voltage at which partial discharges can be detected. In the case of power frequency, measurements are made according to IEC/TS 60034-27, within the limit of sensitivity described in 9.4. In the case of the PD measurement by impulse voltage, tests are performed according to IEC/TS 61934, which describes the sensitivity checks and reporting requirements.

Successful service experience may have enabled a manufacturer to assign a stress category to a machine using a specific design of insulation system. In this case and with the agreement of the purchaser, this experience may be used as an alternative way to qualify the insulation system for converter-fed applications.

10.2 Approach

10.2.1 General

Type I insulation systems are intended for use in the absence of partial discharge activity throughout their lifetime. It is essential to measure whether partial discharge occurs at the specified test voltage (see Clause B.6). The allowable voltage waveshapes for the different insulation system components are shown in Table 5 and Figures B.1 and B.2. The test samples are to be made using the materials and manufacturing methods applicable to production. They may represent either part of the insulation system or the insulation system in total.

10.2.2 Twisted pair or equivalent arrangement

Twisted pairs or parallel adjacent conductors, in the case of flat wires, may be used for testing turn/turn behaviour in order to establish the voltage stress level to be used between parallel conductors during the motorette/formette test. Sine and impulse voltages are assumed to give equivalent results in terms of PD inception voltage because the voltage distribution experienced in a complete winding is not represented. Consequently, either voltage waveform may be used in testing for PD inception. The voltage stress shall be raised above the average stress per turn expected in service to account for the stress concentration described in 5.4. The test levels specified in Annex B are peak/peak.

10.2.3 Motorette (random wound) or formette (form-wound)

These models may be used to represent phase/phase and phase/ground insulation. Turn/turn insulation may also be represented by the use of parallel conductors, when sinewave voltage may be used. This model arrangement should be tested according to Table 5. The applied voltages stressing the model insulation components should reproduce the voltage stresses occurring within the complete machine in service.

10.2.4 Complete windings

This allows phase/phase and phase/ground insulation to be tested. Turn/turn insulation in conventional windings should be tested by the use of impulse voltage. The advantage of an impulse wave shape for testing is that all areas of the winding can be stressed in a representative way, even if the star point is connected. Nevertheless, it is important to take account of the voltage distribution within the winding. Tests should be performed on impregnated samples where the service machine is to have impregnated windings.

When the test object is a complete winding and the capacitance between the winding and ground is large, the rise time of the impulse voltage applied to the test object may become longer than the original rise time of the impulse generator. This phenomenon may occur when

the impulse generator cannot provide enough capacitive current for the test object to generate a steep-fronted voltage rise. It may be overcome by improvements to the current capacity and output impedance of the impulse generator. In recognition of the difficulty in achieving the required rise time when testing large machines using commercially available impulse generators, the tolerance on the rise time of the waveform is relaxed for larger machines as described in Clause B.2.

Sinewave testing of PD activity is allowed if a special winding is prepared in which pairs of wires are used and the maximum voltage that is expected to develop across two neighbouring turns is applied between them. The occasions where sinewave or impulse voltage shall be used are given in Table 5.

Table 5 – Allowable voltage waveforms for testing system components

Component to be tested	Design qualification tests				Type test	
	Twisted pair or equivalent		Motorette or formette or complete winding		Complete winding	
	Sine	Impulse	Sine	Impulse	Sine	Impulse
Turn/turn	✓	✓	*	✓	*	✓
Phase/phase	No	No	✓	✓	✓	✓
Phase/ground	No	No	✓	✓	✓	✓
* A special test winding is required in which the turn/turn insulation is simulated by at least two electrically isolated conductors wound in parallel, one of which is grounded and the other is energised.						

10.3 Preparation of test objects

10.3.1 General

The test samples should be as close as possible to the designs described in IEC 60034-18-21 for random wound samples and IEC 60034-18-31 for form-wound samples or to the production winding.

10.3.2 Turn/turn insulation samples

In the turn/turn test, the PDIV or RPDIV is measured on a simple twisted pair sample (see IEC 60172) or equivalent arrangement. This test may be the basis for establishing the voltage stress level to be used between parallel conductors during the motorette/formette test.

Partial discharge tests on impregnated twisted pair samples may not be used to assess the impregnation resin and manufacturing route used in motorettes or formettes or a complete winding. For that evaluation, only motorettes/formettes or complete windings are allowed.

10.3.3 Motorette/formette test samples or complete windings

The samples are made using the same materials and processes that are applied to service coils. Where phase/phase insulation is to be used, as in the case of two coils of different phases occupying the same slot, the model or complete winding shall reproduce this design feature. Tests should be performed on impregnated samples where the service machine is to have impregnated windings. The phase-to-phase and main wall insulation should replicate all creepage distances and clearances (phase to phase) found within the production winding. The samples should replicate production insulating materials and thicknesses for each type of insulation to be tested. In the case of testing with complete windings, the procedures given in 11.2 and 11.3 shall be followed.

10.4 Design qualification tests

10.4.1 General

The aim is to perform thermal ageing tests on the components and related parts of the insulation system, according to IEC 60034-18-21 or IEC 60034-18-31, and to determine at what point PD inception occurs below the specified test voltage. The stress category for which the system is to be qualified (Clause 7), the ageing procedure and diagnostic data shall be reported at the conclusion of the test. The candidate insulation system design is deemed to be qualified if it endures the same or more ageing cycles (without the PD inception occurring below the specified test values) than a reference system with proven service experience. There shall be at least five samples on which PD tests are performed in order to achieve a statistically valid outcome to the test, except in the case of complete windings where one sample is adequate.

10.4.2 Pre-diagnostic tests

A pre-diagnostic electrical ageing test shall be undertaken for 24 h at ambient temperature in which an elevated frequency is applied, with the voltage appropriate for the chosen stress category enhanced by a factor according to Table B.2. The purpose of this test is to detect at an early stage the presence of any dielectric materials which have high loss tangent and, although performing satisfactorily at power frequencies, may result in overheating at the higher frequencies encountered during converter operation. The choice of frequency is determined by the maximum impulse voltage repetition rate expected in service.

10.4.3 Diagnostic tests

Diagnostic tests shall be performed on samples prior to the first ageing cycle and after each ageing sub-cycle as follows. Each electrical diagnostic test shall be performed on each coil of each motorette/formette or winding component. In addition to the diagnostic tests specified in IEC 60034-18-21 or IEC 60034-18-31, a PD test shall be undertaken to check whether PD inception has occurred at the specified test level (Table 4 and Clause B.1).

10.4.4 Ageing cycle

The appropriate thermal ageing sub-cycles specified in IEC 60034-18-21 or IEC 60034-18-31 should be applied to the test samples in order to qualify the systems for the required thermal class. If the thermal class for the insulation system under test has already been determined through testing according to IEC 60034-18-21 and IEC 60034-18-31, it is only necessary to perform ageing at one of the appropriate temperatures. The specified diagnostic sub-cycles in IEC 60034-18-21 and IEC 60034-18-31 shall also be applied. These shall include the mechanical, moisture and voltage tests. It should be noted that moisture may affect the PD test results. For this reason, the moisture test shall be performed at a different stage of the procedure so that the PD test at the end does not follow it immediately.

10.4.5 PD tests

With regard to the PD tests, they should be undertaken:

- a) Phase/phase.
- b) Phase/ground.
- c) Turn/turn.

It is only necessary for the voltage in the PD test to be raised to the test value and to ensure that PD inception has not taken place. It is not necessary for the inception voltage to be measured. During these tests, all sensors embedded within the winding shall be connected electrically to the stator core.

For the phase/ground insulation, the test shall be performed at each phase terminal to ground consecutively, with the other phase terminals and the core grounded. For phase/phase

insulation, the test voltage should be applied in turn between each terminal and the other two with the core floating. In the case of sinewave AC voltage application, it will be necessary to disconnect each phase winding before this test. Each of the three possible combinations shall be tested in turn. The turn/turn test may be undertaken with sinewave AC voltage when parallel conductors are used. Otherwise, the voltage stress between turns is generated by means of impulse voltage test equipment. It will be necessary to detect the partial discharge activity during these tests with measuring equipment capable of operating under impulse conditions (see IEC/TS 61934).

A visual inspection of the test samples to observe the condition of the insulation materials should be performed and reported but this is not an end-point criterion for evaluation.

10.5 Pass criterion for the design qualification test

The pass criterion for qualification of a Type I test sample is that partial discharge inception is above the operating voltage for the selected stress category voltage (Tables B.3 and B.4), multiplied by an enhancement factor (Table B.2), with testing performed on samples which have had thermal ageing. The candidate system shall endure the same or more ageing cycles than the reference system without the PD inception occurring below the specified value. An example of how to calculate the test voltage is described in Clause B.6.

Where partial discharge tests are performed under impulse conditions, the impulse voltage shall conform to the requirements described in Clause B.2.

11 Type test procedure for Type I insulation systems

11.1 General

The type tests are performed on a complete winding or machine. Acceptance is achieved through a series of tests in which the PD inception voltage shall exceed specified values. Tests are performed under impulse conditions or at power frequency, according to Table 5. If the design qualification testing has been performed successfully on a complete winding, the insulation system has effectively passed its type test and no additional type test is required.

The testing with sinusoidal voltage at power frequency when all phases can be separated is more easily performed and leads to conservative test results. The impulse testing is a more realistic representation of the expected stress under converter. In addition, an impulsive voltage waveform as close as possible to the real one provided by the converter shall be available for the test. It shall in any case conform to Figures B.1 and B.2a. The results may differ due to the different test voltage distributions within the winding under test [2].

For sinewave testing, PD inception is recognised when there is at least 1 pulse detected per cycle. For impulse testing, the equivalent test is for there to be at least one pulse detected during all the impulses which occur in one fundamental operating cycle.

Partial discharge testing is performed at the maximum peak/peak operating voltage multiplied by an enhancement factor. An example of how to calculate the test voltage is described in Clause B.6. The temperature of the winding shall be $20\text{ }^{\circ}\text{C} \pm 10\text{ }^{\circ}\text{C}$. This test is performed by agreement between the purchaser and manufacturer.

11.2 Power frequency PD tests

The complete winding is first of all subjected to partial discharge tests at 50 Hz/60 Hz as follows. The operating voltage levels given in Table B.3 are increased by the relevant enhancement factor in Table B.2. If the partial discharge inception voltage is below the test voltage, the winding has failed.

- a) A phase/ground PD test is performed between all the phase terminals and the core. It is not necessary to disconnect any windings.

- b) A phase/phase PD test is performed with the core floating and the star point disconnected. For delta windings, all the phases should be disconnected and the core floating. If the star/delta point cannot be disconnected to isolate the phases, impulse testing shall be used.

11.3 Impulse PD tests

Impulse testing is more representative of service [4][5][6]. The impulse shape may influence the test result and therefore an impulsive voltage waveform as close as possible to the one provided by the converter in service shall be used for the test. If the repetitive partial discharge inception voltage is below the specified test voltage, the winding has failed.

- a) Perform a phase/ground PD test with impulse voltage having the expected rise time from the converter and with the specified waveform according to Figure B.2. The rise time can be longer than the specified one as it is less significant according to Table 3. With a longer rise time, overstressing of the turn/turn insulation is reduced. The test should be performed at each phase terminal to ground consecutively, with the other phase terminals and the core grounded. Note that the impulse test voltage is applied not only to the phase/ground insulation but also to the phase/phase insulation. Therefore, if the specified phase/ground test voltage is higher than the phase/phase one, the phase/ground insulation shall be tested using power frequency voltage so that no stress occurs on the phase/phase insulation.
- b) Perform a phase/phase PD test with impulse voltage of the specified waveform according to Figure B.2. The impulse test may only be performed if no PD inception has been detected during the phase/ground insulation test. The rise time may be longer than the specified one as it is less significant according to Table 3. In this way, overstressing the turn/turn insulation can be reduced. The test voltage should be applied in turn between each terminal and the other two with the core floating. In this way, overstressing the phase/ground insulation is reduced.
- c) Perform a turn/turn insulation test by applying impulse voltages between each phase terminal and ground consecutively, with the other phase terminals and the core grounded. The impulse voltage rise time shall be in accordance with the values given in Table B.1 and Figures B.1 and B.2 and the amplitude in accordance with Clause B.6. If the RPDIV is below the pass criterion, the winding has failed. If the RPDIV is above the pass criterion, the winding has passed the test. If the machine is to be assigned a special impulse voltage insulation class for the turn/turn insulation (see Clause C.2), this test is performed using parallel conductors in the winding if no other way is possible.

12 Routine tests

It is good practice to perform a routine test on each winding. For this purpose, it is recommended that the tests described in 11.2 and 11.3 are performed in agreement between the manufacturer and customer.

13 Analysis, reporting and classification

The approach given in 6.2 of IEC 60034-18-1:2010 to analysis, reporting and classification should be adopted so that all relevant data is analysed correctly and reported in a traceable manner. In particular, a photographic record of the impulse voltage waveform at the machine terminals during tests shall be provided.

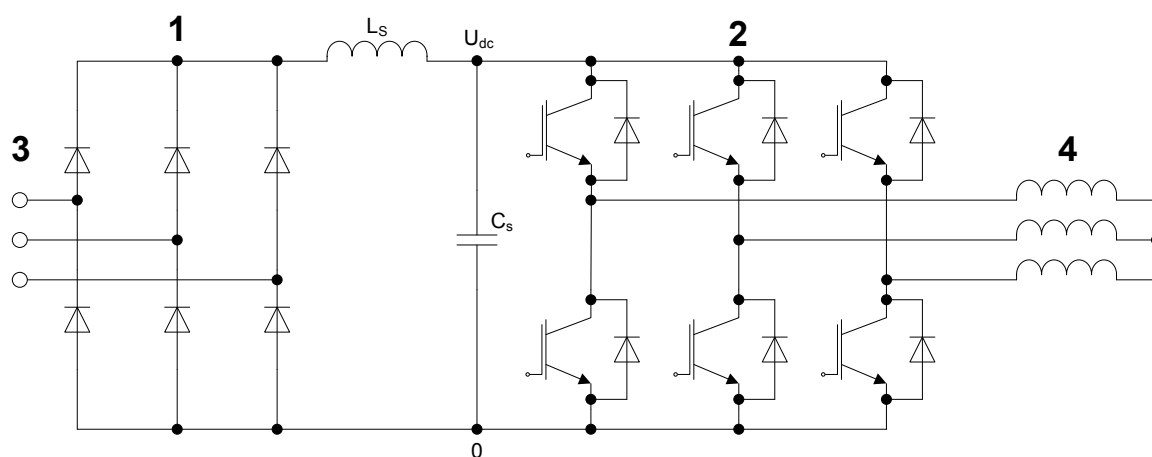
Annex A (informative)

Derivation of possible terminal voltages in service for a converter-fed machine

A.1 Calculation of d.c. bus voltage

A U-converter, pulse width modulation (PWM) or vector machine power control generates rectangular waves of fixed amplitude voltage that have varying width and repetition rate. The voltage of the impulses at the output of the converter is not more than the d.c. bus voltage (U_{dc}). This voltage depends on the rectified mains voltage or braking voltage level or power factor correction regulation voltage.

A simplified circuit diagram for a converter/machine system is shown in Figure A.1.



IEC 0568/14

Key

- 1 Rectifier
- 2 PWM converter
- 3 AC input
- 4 Machine winding

L_s and C_s are system inductance and capacitance

Figure A.1 – Circuit diagram for a converter/machine system

As a first step, it is necessary to define a neutral point from which all voltages may be referenced. For example, in a 3-phase system:

$$u_1(t) = \sqrt{2}U_{\text{phase}} \sin(\omega t)$$

$$u_2(t) = \sqrt{2}U_{\text{phase}} \sin\left(\omega t - \frac{2\pi}{3}\right)$$

$$u_3(t) = \sqrt{2}U_{\text{phase}} \sin\left(\omega t - \frac{4\pi}{3}\right)$$

This may be extended to a multiphase system. The general law for defining the magnitude of the voltage of a multiphase a.c. voltage after rectification is given by:

$$U_{dc} = a \left(\frac{\varphi}{\pi} \right) \sin \left(\frac{\pi}{\varphi} \right) U_{max} = k \cdot U_{max}$$

Where:

k is a constant;

a is the rectifying type ($a = 1$ for half wave rectification or $a = 2$ for full wave rectification);

φ is the number of phases (minimum of 2).

The formula only applies to balanced systems and so φ has a minimum value of 2. In the special case of a single-phase drive, it is necessary to consider the voltage to be a combination of two voltages, each $\sqrt{2}U/2$, with $\varphi = 2$.

The voltage at the output of the d.c. bus (U_{dc}) and therefore the peak value of the output voltage of the converter may be calculated using the above formula. For example, in the general case of a three-phase sinusoidal voltage fully rectified it leads to

$$U_{dc} = 2 \cdot \frac{3}{\pi} \cdot \sin \left(\frac{\pi}{3} \right) \cdot \sqrt{2}U_{phase} = 2 \cdot \frac{3}{\pi} \cdot \frac{\sqrt{3}}{2} \cdot \sqrt{2}U_{phase} = \frac{3}{\pi} \cdot \sqrt{3} \sqrt{2}U_{phase} \text{ and}$$

$$\sqrt{3}U_{phase} = U_{line} \text{ (that is, the r.m.s. phase to phase voltage).}$$

$$\text{Therefore, } U_{dc} = \frac{3}{\pi} \cdot \sqrt{2}U_{line}$$

For example, if $U_{line} = 500 \text{ V r.m.s.}$ then

$$U_{dc} = 500 \frac{3}{\pi} \sqrt{2} = 675 \text{ V} = 1,35U_{line} = 1,65U_{max}$$

The factor of 1,35 has been derived from theoretical considerations but in practice may be larger. For example, in cases of regeneration, the DC bus voltage maximum limit is the cut-off voltage in the drive, which will be much higher than the mains derived DC bus voltage.

A.2 Calculation of maximum peak voltages for a 2-level converter

The peak voltages are a consequence of the cable and machine arrangements, which result in overshoot voltages of $0 U_{dc}$, $0,5 U_{dc}$, U_{dc} or $1,5 U_{dc}$ superimposed on the converter output voltage. In order to allow for practical considerations, the value of overshoot factor for the lowest stress category is raised to 1,1. The values shown in Table A.1 arise from the following calculation.

$$U_{dc} = \text{DC link voltage} = \frac{3}{\pi} \sqrt{2}U_{line} = \frac{3}{\pi} \sqrt{3}U_{max}$$

Phase/phase voltage on the machine = DC link voltage $\times$ overshoot factor (1,1, 1,5, 2 or 2,5).

To convert to units of U_{max} , the voltage is divided by the maximum phase/neutral voltage for the system as calculated in Clause A.1.

Table A.1 – Examples of maximum peak voltages

Rated voltage $U_{r.m.s.}$	U_{dc} V	Overshoot factor U_p/U_a	U_p V	U_{max} V	$U_{p/p} / U_{max}$
500	675	1,1	743	409	1,82
500	675	1,5	1 013	409	2,48
500	675	2	1 350	409	3,30
500	675	2,5	1 688	409	4,13

Annex B (normative)

Derivation of test voltages for Type I insulation systems

B.1 Stress categories

The test voltages to be used in this standard for qualification and type testing of an insulation system are derived from the stress category that has been selected. The stress categories given in Clause 7 provide operating ranges for the peak voltage appearing at the terminals of a rotating machine driven from a 2-level converter and the default rise time of the impulses. A careful selection shall therefore be made of the specified tests so that appropriate testing is performed on the insulation system. The stress categories are shown again for convenience in Table B.1.

Table B.1 – Summary of stress categories

Stress category	Overshoot factor (OF) U_p/U_a	Impulse rise time t_r μs
A – Benign	$OF \leq 1,1$	0,3
B – Moderate	$1,1 < OF \leq 1,5$	
C – Severe	$1,5 < OF \leq 2,0$	
D – Extreme	$2,0 < OF \leq 2,5$	

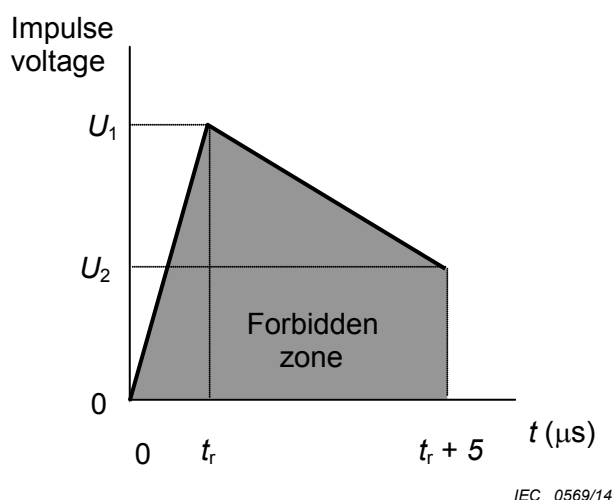
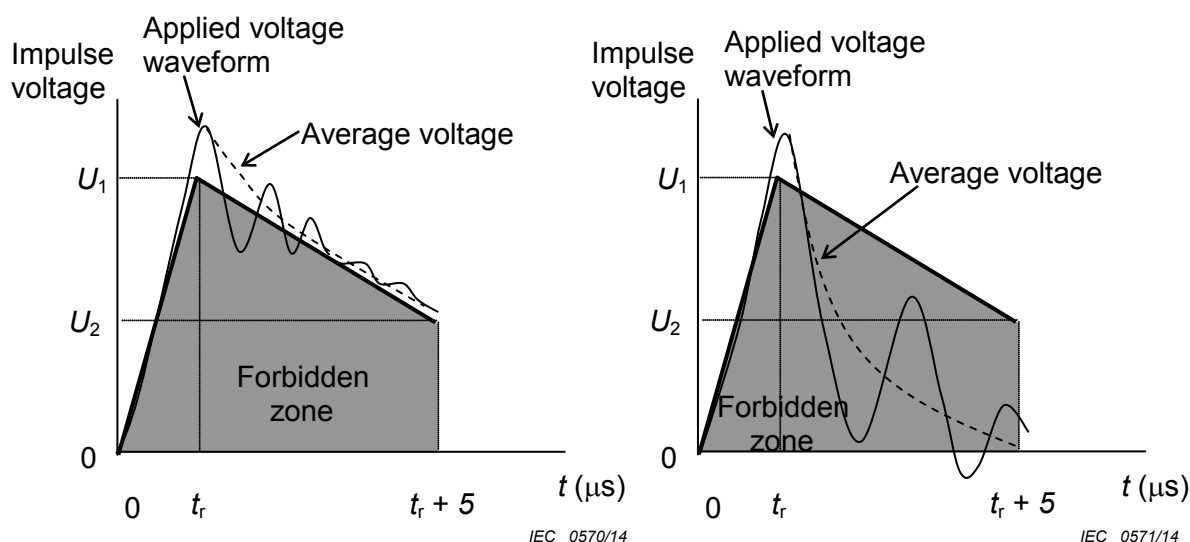
B.2 Requirements for the applied impulse voltage

The overall width of the impulse to be used in performing the RPDIV tests shall be long enough to allow initiation of a PD pulse [5]. The impulse waveshape to be used consists of a rising front with rise time 0,3 μs . The average peak voltage shall gradually decrease to be equal to, or exceed the slope from U_1 to U_2 until a time of 5 μs has elapsed. This is explained in terms of a forbidden zone which the applied impulse should not enter, as shown in Figure B.1. In recognition of the difficulty which may occur in achieving the required rise time using commercially available impulse generators, the tolerance on the rise time of the waveform is relaxed to $\pm 0,2 \mu s$.

It is accepted that, in practice, it may not be possible to achieve precise values for the impulse waveshape and the following deviations are permitted.

- a) The start and end of the rising front are permitted to deviate from linearity by 10 %.
- b) Voltage oscillation is permitted at the peak of the impulse voltage, provided the average voltage does not enter the forbidden zone.
- c) The peak of the impulse voltage shall not be less than 97 % of the value appropriate for the chosen stress category.

Examples of acceptable and unacceptable impulse voltage waveforms are shown in Figure B.2.

**Key** U_1 Test voltage $U_2 = U_1 \times U_a/U_p$ **Figure B.1 – Forbidden zone (shaded) for impulse tests****Figure B.2a – Acceptable waveform for the voltage used in the impulse PD test (key as in Figure B.1)****Figure B.2b – Unacceptable waveform for the voltage used in the impulse PD test****Figure B.2 – Examples of test waveforms****B.3 Enhancement factors for PD tests**

There are various factors that should be considered for enhancement of the maximum peak/peak operating voltages specified in the qualification and type tests for Type I insulation systems. The first arises from the requirements of IEC 60664-1, which lists safety factors by which the PD test voltages should be increased. A safety factor of 1,25 relates to the hysteresis effect by which PDEV is said from practical experience to be 25 % below the PDIV. This is relevant if PD activity is initiated by a transient overvoltage exceeding the normal peak voltage.

A second enhancement factor to be considered arises from the claim in 7.6 of IEC/TS 60034-25:2007 that an increase in winding temperature from 25 °C to 155 °C typically results in a fall in the PDIV by 30 %. For phase to ground testing, this factor is reduced due to the cooling effect of the slot on the neighbouring turns and only adds a further 5 % – 10 %.

Finally, the effect of thermal ageing should be considered. If the machine is being operated near to the class temperature of the insulation, it is reasonable to expect PD inception voltages to fall as a result of thermal ageing in service. It is a requirement for Type I machines that they remain PD free for their lifetime. An empirical factor should therefore be introduced to raise the test voltages on new machines to take account of the effect of ageing. A change in service application may also change the ageing rate. A value of up to 1,2 is proposed but a smaller value may be used, based on service experience or laboratory testing. The precise value of the ageing factor depends upon the service temperature and if this were significantly below the temperature class of the insulation, it may be unnecessary to apply it. The ageing factor has a minimum value of 1,0. For a service temperature near to the class temperature, the following formula may be used.

$$\text{Ageing factor} = 1,2 \left(1 - \frac{\text{Class temperature} - \text{Service temperature}}{\text{Class temperature}} \right)$$

A summary of the enhancement factors to be applied to the operating voltages is given in Table B.2.

Table B.2 – Summary of enhancement factors to be applied to the operating voltages

		Enhancement factors (EF)			
		PD Safety factor	Temperature	Ageing	Total EF
Qualification test	Phase/phase	1,25	1,0 – 1,3	1,0	1,25 – 1,63
	Phase/ground		1,0 – 1,1		1,25 – 1,38
	Turn/turn		1,0 – 1,3		1,25 – 1,63
Type test	Phase/phase	1,25	1,0 – 1,3	1,0 – 1,2	1,25 – 1,95
	Phase/ground		1,0 – 1,1		1,25 – 1,65
	Turn/turn		1,0 – 1,3		1,25 – 1,95
NOTE The total enhancement factor (Column 6) is the product of the individual factors in columns 3-5.					

B.4 Voltage for design qualification and type tests

For the design qualification and type testing of Type I insulation systems, the voltage to be used for PD tests is calculated using the maximum operating voltage for the chosen stress category as shown in Tables B.3 and B.4 for a 2-level converter. Note that the values for U_p and U_b relate to phase/phase voltages on the machine terminals.

The important phase/phase and phase/ground voltages appearing at the terminals of a rotating machine fed from a 2-level converter and the turn/turn voltages of coils of the rotating machine fed from the converter are shown schematically in Figure B.3. Note that $U_{pk/pk}$ is the main stressing voltage of the phase/phase, phase/ground, and turn/turn insulation because the effects of switching impulses are smaller and may be ignored. The jump voltage U_j of the phase/ground voltage is important for turn/turn insulation because it generates additional